Chapter 14 · Semester 2

CAPACITORS

Notes by Cikgu Suhaili

15 worked examples 5 simulations Exercises with answers

14.1 Capacitance

- A **capacitor** is a device that stores electric charge (and electrical energy).
- It is made of two conducting plates separated by a small gap of air or a thin insulator called a **dielectric** (mica, ceramic, paper, plastic, oil).

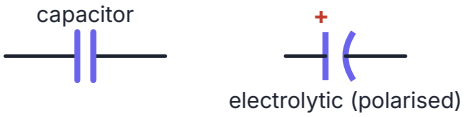


Figure 14.1: circuit symbols for a capacitor.

Capacitance, C

the **ratio of the charge on either plate to the potential difference between the plates**.

$$C = \frac{Q}{V}$$

Q = charge on one plate (the other has $-Q$); V = p.d. across the plates. Unit: farad (F) = C V^{-1}

- 1 F is the capacitance when a charge of 1 C on each plate gives a p.d. of 1 V.
- The farad is a very large unit: 1 μF = 10^{-6} F, 1 nF = 10^{-9} F, 1 pF = 10^{-12} F.
- For a given capacitor C is constant, so $Q \propto V$.
- The net charge on a capacitor is zero: it holds $+Q$ on one plate and $-Q$ on the other. "The charge on a capacitor" means Q.

14.2 Parallel plate capacitors

How a parallel plate capacitor is charged

- When the capacitor is connected to a battery, electrons flow from the negative terminal of the battery to one plate, which becomes negatively charged ($-Q$).
- At the same time, electrons are pulled off the other plate towards the positive terminal, leaving it positively charged ($+Q$). No charge crosses the gap.
- As charge builds up, the p.d. across the plates rises. The flow of electrons stops when the p.d. across the capacitor equals the e.m.f. of the battery.

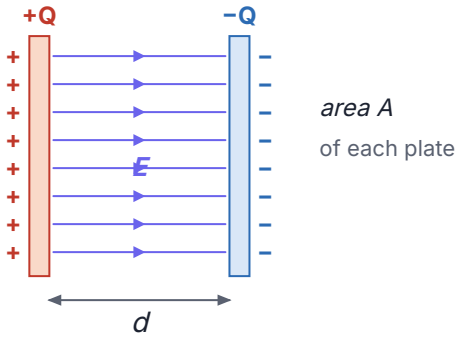


Figure 14.2: a charged parallel plate capacitor. The field between the plates is uniform.

Deriving C = ε₀A/d

Gauss's law gives the field between the plates: $E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}$

The field is uniform, so the p.d. is $V = Ed = \frac{Qd}{\varepsilon_0 A}$

$$C = \frac{Q}{V} = \frac{Q}{Qd/\varepsilon_0 A} \Rightarrow C_0 = \frac{\varepsilon_0 A}{d}$$

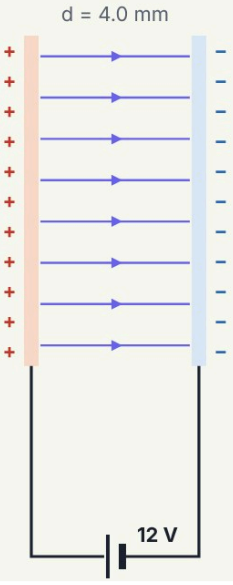
$$C_0 = \frac{\varepsilon_0 A}{d}$$

A = area of overlap of the plates; d = separation; $\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1}. The capacitance is proportional to A and inversely proportional to d.

Simulation

Parallel plate capacitor

Change the area, the separation and the dielectric. Compare a capacitor that stays connected to the battery (V fixed) with one that is disconnected first (Q fixed).



capacitance C	charge Q	p.d. V	field E = V/d	energy U = ½QV
4.43 × 10⁻¹¹ F	**5.31 × 10⁻¹⁰ C**	**12.00 V**	**3.00 × 10³ V m⁻¹**	**3.19 × 10⁻⁹ J**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 14 to use the simulation.

Example 14.2.1

(a) Find the capacitance of a capacitor whose plates are 20 cm × 3.0 cm, separated by a 1.0 mm air gap. (b) What is the charge on each plate when it is connected to a 12 V battery? (c) What is the electric field between the plates?

Solution

(a) $C = \dfrac{\varepsilon_0 A}{d} = \dfrac{(8.85 \times 10^{-12})(0.20 \times 0.030)}{1.0 \times 10^{-3}} = \mathbf{5.3 \times 10^{-11}\ F}$ (53 pF)

(b) $Q = CV = (5.3 \times 10^{-11})(12) = \mathbf{6.4 \times 10^{-10}\ C}$

(c) $E = \dfrac{V}{d} = \dfrac{12}{1.0 \times 10^{-3}} = \mathbf{1.2 \times 10^{4}\ V\ m^{-1}}$

Example 14.2.2

An electric field of 2.80×10^5 V m^{-1} is needed between two parallel plates, each of area 21.0 cm^2 and 0.250 cm apart in air. Find the charge on each plate.

Solution

$$Q = CV = \frac{\varepsilon_0 A}{d}(Ed) = \varepsilon_0 AE$$

$Q = (8.85 \times 10^{-12})(21.0 \times 10^{-4})(2.80 \times 10^5) =$ **5.20×10^{-9} C** (the separation is not needed)

Exercise: parallel plate capacitors

1. The plates of a parallel plate capacitor in vacuum are 8.0 mm apart and each has an area of 4.0 cm^2. The p.d. across the plates is 2.0 kV. Find (a) the capacitance, (b) the charge on each plate, (c) the electric field between the plates.

   Answer

   (a) 4.4×10^{-13} F (0.44 pF) (b) 8.9×10^{-10} C (c) 2.5×10^5 V m^{-1}

   FULL SOLUTION

   A = 4.0 cm^2 = 4.0×10^{-4} m^2, d = 8.0×10^{-3} m, V = 2.0×10^3 V

   (a) C = ε_0A/d = $(8.85 \times 10^{-12})(4.0 \times 10^{-4}) / (8.0 \times 10^{-3})$

   = **4.4×10^{-13} F** (0.44 pF)

   (b) Q = CV = $(4.43 \times 10^{-13})(2.0 \times 10^3)$

   = **8.9×10^{-10} C**

   (c) E = V/d = $2.0 \times 10^3 / (8.0 \times 10^{-3})$

   = **2.5×10^5 V m^{-1}**

2. The separation of the plates of an isolated charged capacitor is doubled. What happens to (a) C, (b) Q, (c) V?

   Answer

   (a) C halves (b) Q is unchanged (isolated) (c) $V = Q/C$ doubles

   FULL SOLUTION

   (a) C = ε_0A/d: doubling d makes **C halve**.

   (b) The capacitor is isolated, so no charge can flow on or off: **Q is unchanged**.

   (c) V = Q/C: Q is constant and C halves, so **V doubles**.

14.3 Dielectrics

- A **dielectric** is an insulator: it has no free electrons.
- When a dielectric fills the space between the plates, the capacitance increases by a factor ε_r, the **dielectric constant** (relative permittivity) of the material.

Dielectric constant, ε_r

the **ratio of the capacitance of a capacitor with the dielectric between the plates to its capacitance with a vacuum between the plates**.

$$\varepsilon_r = \frac{C}{C_0} \qquad C = \frac{\varepsilon A}{d} = \frac{\varepsilon_r \varepsilon_0 A}{d}$$

$\varepsilon = \varepsilon_r \varepsilon_0$ is the permittivity of the dielectric; ε_r has no unit and is always greater than 1

Why a dielectric increases the capacitance

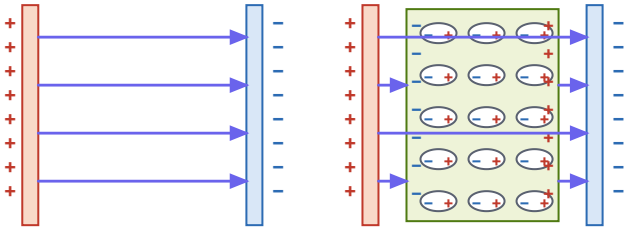


Figure 14.3: the molecules of the dielectric are polarised; induced surface charges appear on its faces.

The electric field between the plates **polarises** the molecules of the dielectric. In a **polar** dielectric (e.g. water) the permanent dipoles line up with the field; in a **non-polar** dielectric the field pulls the centres of positive and negative charge slightly apart, making induced dipoles.

The negative ends face the positive plate and the positive ends face the negative plate, so **induced surface charges** appear: negative on the face next to the + plate, positive on the face next to the − plate.

These surface charges produce a field opposite to the field of the plates, so the resultant field in the dielectric is smaller: $E = E_0/\varepsilon_r$.

For the same charge Q, the p.d. $V = Ed$ is smaller, so $C = Q/V$ is larger.

$$\varepsilon_r = \frac{C}{C_0} = \frac{V_0}{V} = \frac{E_0}{E} \quad \text{(same charge } Q\text{)}$$

Quantity	Isolated capacitor (Q fixed), dielectric inserted	Capacitor kept connected to the battery (V fixed)
Capacitance	$C = \varepsilon_r C_0$ (increases)	$C = \varepsilon_r C_0$ (increases)
Charge	unchanged	$Q = \varepsilon_r Q_0$ (the battery supplies more)
p.d.	$V = V_0/\varepsilon_r$ (decreases)	unchanged
Field	$E = E_0/\varepsilon_r$ (decreases)	unchanged ($E = V/d$)
Energy stored	$U = Q^2/2C$: decreases	$U = \frac{1}{2}CV^2$: increases

Material	Dielectric constant ε_r	Dielectric strength / 10^6 V m^{-1}
Air	1.0006	3
Teflon	2.1	60
Silicone oil	2.5	15
Mylar	3.2	7
Paper	3.7	16
Water	80	—

The **dielectric strength** is the largest electric field a material can stand before it breaks down and conducts. A dielectric also lets the plates be very close together without touching.

Example 14.3.1

A parallel plate capacitor has plates of area $2.0 \times 10^{-2}\ \text{m}^2$, 1.0 cm apart in air. It is charged to 3000 V and the battery is disconnected. A dielectric sheet is then slid in to fill the gap, and the p.d. falls to 1000 V. Find (a) the capacitance in air, (b) the charge on each plate, (c) the capacitance with the dielectric, (d) the dielectric constant, (e) the permittivity of the dielectric, (f) the field before, (g) the field after the dielectric is inserted.

Solution

(a) $C_0 = \dfrac{(8.85 \times 10^{-12})(2.0 \times 10^{-2})}{0.010} = \mathbf{1.77 \times 10^{-11}\ F}$

(b) $Q_0 = C_0 V_0 = (1.77 \times 10^{-11})(3000) = \mathbf{5.31 \times 10^{-8}\ C}$

(c) the charge stays the same: $C = Q_0/V = 5.31 \times 10^{-8}/1000 = \mathbf{5.31 \times 10^{-11}\ F}$

(d) $\varepsilon_r = C/C_0 = \mathbf{3.0}$

(e) $\varepsilon = \varepsilon_r \varepsilon_0 = \mathbf{2.66 \times 10^{-11}\ F\ m^{-1}}$

(f) $E_0 = V_0/d = 3000/0.010 = \mathbf{3.0 \times 10^{5}\ V\ m^{-1}}$

(g) $E = E_0/\varepsilon_r = \mathbf{1.0 \times 10^{5}\ V\ m^{-1}}$

Teacher's note

First ask: is the capacitor still connected to the battery? If yes, V stays the same. If it was disconnected first, Q stays the same. Everything else follows from $C = Q/V$.

A natural capacitor: the thundercloud and the ground

The base of a thundercloud and the ground below it behave like the two plates of a huge parallel plate capacitor, with the **air between them as the dielectric**.

Inside the cloud, collisions between ice crystals and soft hail separate charge. The base of the cloud usually becomes **negative**.

This negative charge repels electrons in the ground below, so the ground surface becomes **positive** (charging by induction): the two "plates" carry $+Q$ and $-Q$.

As more charge builds up, the electric field in the air between them grows: $E = V/d = Q/(\varepsilon_0 A)$.

When E reaches the **dielectric strength of air** (about 3×10^6 V m^{-1}), the air is ionised and becomes a conductor. The dielectric breaks down and the capacitor discharges through it: this is **lightning**.

The huge current heats the air channel to about 30 000 K. The air expands suddenly, and we hear this as **thunder**. We see the flash almost at once (light travels at 3×10^8 m s^{-1}) but hear the thunder later (sound travels at about 340 m s^{-1}): distance ≈ 340 × delay, roughly 1 km for every 3 s.

Simulation

Thundercloud and ground: a natural capacitor

Let the storm charge the cloud. Watch the charges on the cloud base and the ground, and the field in the air between them. When the field reaches the dielectric strength of air, the air breaks down and the capacitor discharges as lightning. Try a higher cloud: does it flash at a different charge? Turn on the thunder sound and count the seconds between the flash and the thunder. Use slow motion to see the stepped leader come down before the bright return stroke.



charge on the cloud Q	field E = Q/(ε₀A)	capacitance C = ε₀A/d	p.d. V = Q/C
10.8 C	**1.22 × 10⁶ V m⁻¹**	**8.8 nF**	**1.22 × 10⁹ V**

energy stored ½QV	flashes	thunder heard after d/340
6.6 × 10⁹ J	**0**	**2.9 s**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 14 to use the simulation.

Example 14.3.2

The base of a thundercloud is 1.0 km above flat ground and has an area of 1.0 km^2. Treat the cloud and the ground as a parallel plate capacitor with air between them (dielectric strength 3.0×10^6 V m^{-1}). Find (a) the capacitance, (b) the largest charge the cloud can hold before the air breaks down, (c) the p.d. between the cloud and the ground at that moment, (d) the energy released if the whole charge is discharged.

Solution

(a) $C = \dfrac{\varepsilon_0 A}{d} = \dfrac{(8.85 \times 10^{-12})(1.0 \times 10^6)}{1.0 \times 10^3} = \mathbf{8.9 \times 10^{-9}\ F}$ (8.9 nF)

(b) $E = \dfrac{Q}{\varepsilon_0 A} \Rightarrow Q = \varepsilon_0 A E = (8.85 \times 10^{-12})(1.0 \times 10^6)(3.0 \times 10^6) = \mathbf{27\ C}$

(c) $V = Ed = (3.0 \times 10^6)(1.0 \times 10^3) = \mathbf{3.0 \times 10^9\ V}$

(d) $U = \tfrac{1}{2}QV = \tfrac{1}{2}(26.6)(3.0 \times 10^9) = \mathbf{4.0 \times 10^{10}\ J}$

This is an idealised model. In a real storm the flash starts at a lower average field, because the field is much stronger near sharp points, raindrops and ice, and the charge is spread through the cloud. Still, the answers are the right size: a real flash carries tens of coulombs.

14.4 Capacitors in series and in parallel

Series

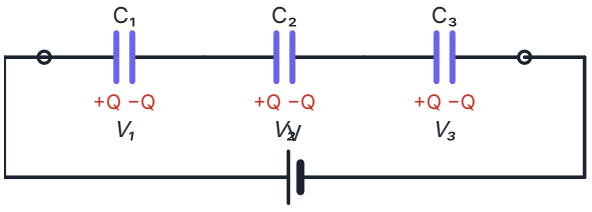


Figure 14.4: three capacitors in series.

Electrons from the battery charge the end plate $-Q$. This repels electrons from the facing plate, which becomes $+Q$, and so on along the chain: **every capacitor carries the same charge Q.**

The p.d.s add up: $V = V_1 + V_2 + V_3 = \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3}$

Divide by Q, with $C_{eq} = Q/V$: $\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$

Parallel

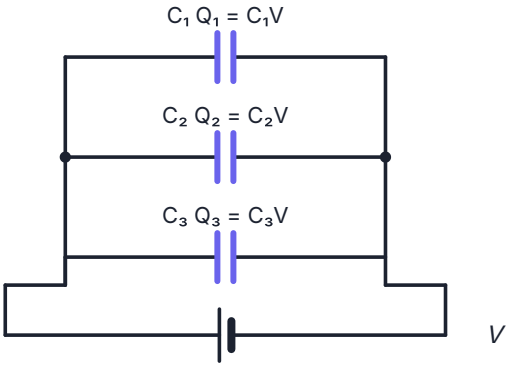


Figure 14.5: three capacitors in parallel.

Each capacitor is connected across the same two points: **every capacitor has the same p.d.** V.

The charges add up: $Q = Q_1 + Q_2 + Q_3 = C_1V + C_2V + C_3V$

Divide by V: $C_{eq} = C_1 + C_2 + C_3$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$$

series: same Q; C_{eq} is smaller than the smallest

$$C_{eq} = C_1 + C_2 + \dots$$

parallel: same V; C_{eq} is larger than the largest

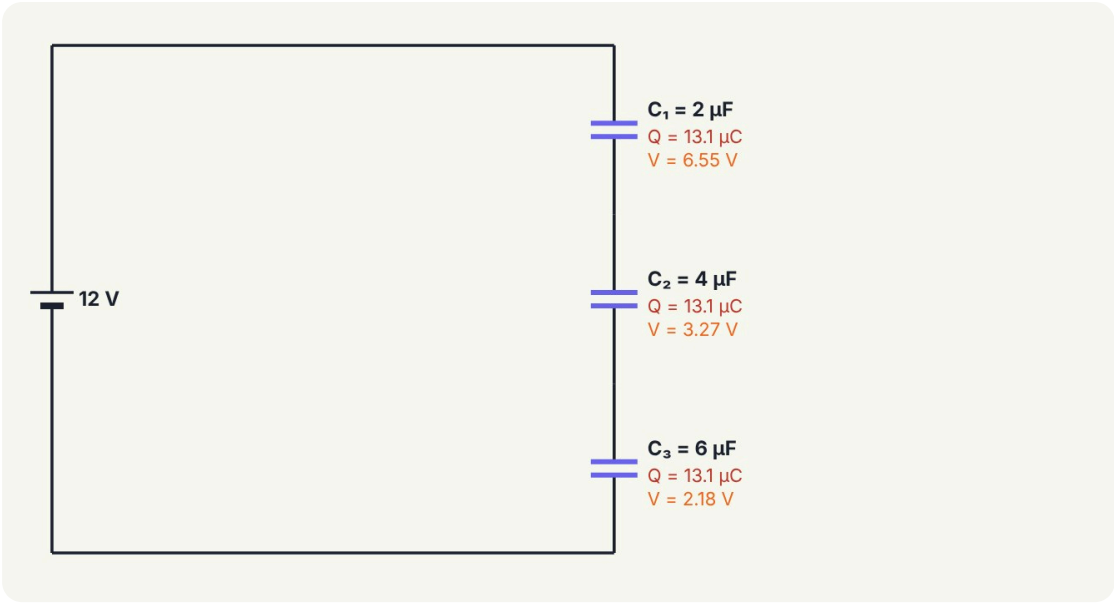
Teacher's note

Two in series: $C_{eq} = \frac{C_1C_2}{C_1 + C_2}$ ("product over sum"). Capacitors combine the opposite way to resistors.

Simulation

Capacitors in series and parallel

Pick an arrangement and change the capacitances and the supply voltage. Check which quantity is shared (Q in series, V in parallel) and how the charge and p.d. split.

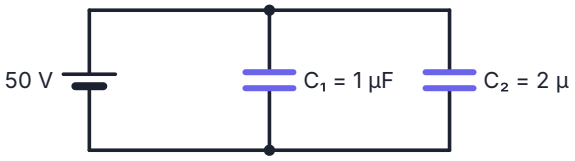


equivalent capacitance	total charge from the supply	energy stored	shared quantity
1.09 µF	**13.1 µC**	**78.5 µJ**	**same Q on each**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 14 to use the simulation.

Example 14.4.1

A 1 µF and a 2 µF capacitor are connected in parallel to a 50 V supply. Find (a) the charge on each capacitor, (b) the equivalent capacitance.



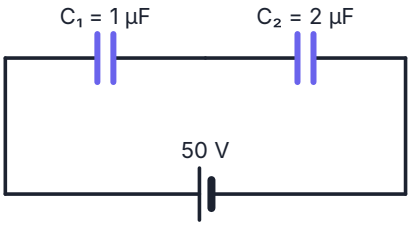
Solution

(a) $Q_1 = C_1V = (1\ \mu F)(50) =$ **50 µC**; $Q_2 = (2\ \mu F)(50) =$ **100 µC**

(b) $C_{eq} = 1 + 2 =$ **3 µF**

Example 14.4.2

The same capacitors are connected in series to the 50 V supply. Find (a) the equivalent capacitance, (b) the charge on each capacitor, (c) the p.d. across each.



Solution

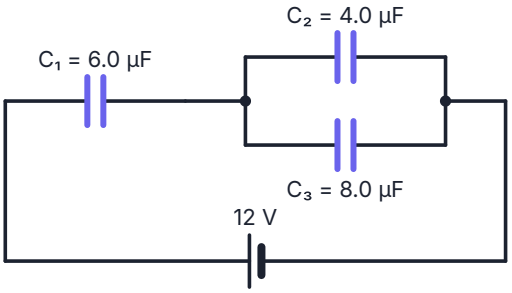
(a) $\frac{1}{C_{eq}} = \frac{1}{1} + \frac{1}{2} \Rightarrow C_{eq} =$ **0.667 μF**

(b) $Q = C_{eq}V$ = (0.667 μF)(50) = **33.3 μC** on each capacitor

(c) $V_1 = Q/C_1$ = 33.3/1 = **33.3 V**; V_2 = 33.3/2 = **16.7 V** (check: 33.3 + 16.7 = 50 V)

Example 14.4.3

Find (a) the equivalent capacitance, (b) the charge on each capacitor, (c) the p.d. across each capacitor.



Solution

(a) C_{23} = 4.0 + 8.0 = 12.0 μF; $\frac{1}{C_{eq}} = \frac{1}{6.0} + \frac{1}{12.0} \Rightarrow C_{eq} =$ **4.0 μF**

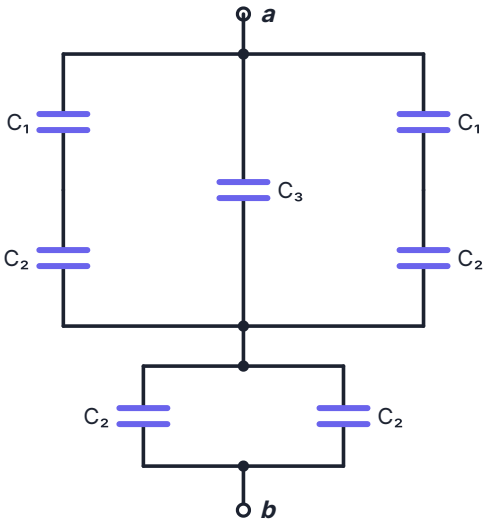
(b) $Q_1 = Q_{total}$ = (4.0 μF)(12) = **48 μC**

(c) V_1 = 48/6.0 = **8.0 V**; $V_2 = V_3$ = 12 − 8.0 = **4.0 V**

(b) Q_2 = (4.0 μF)(4.0) = **16 μC**; Q_3 = (8.0 μF)(4.0) = **32 μC** (check: 16 + 32 = 48 μC)

Example 14.4.4

Find the equivalent capacitance between a and b. C_1 = 5.00 μF, C_2 = 10.0 μF, C_3 = 2.00 μF.



Solution

Each side branch: C_1 and C_2 in series: $C_{12} = \dfrac{(5.00)(10.0)}{15.0} = 3.33\ \mu F$

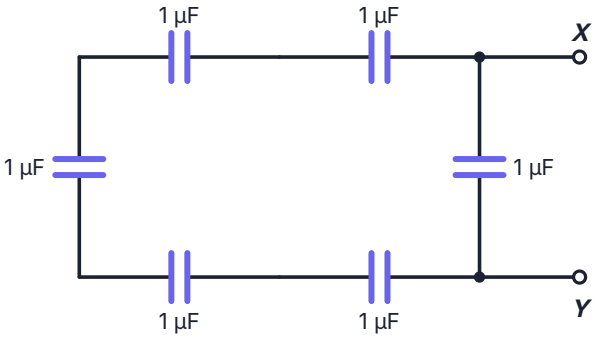
Upper group in parallel: $C_a = 3.33 + 2.00 + 3.33 = 8.67\ \mu F$

Lower pair in parallel: $C_{22} = 10.0 + 10.0 = 20.0\ \mu F$

In series: $\dfrac{1}{C_{eq}} = \dfrac{1}{8.67} + \dfrac{1}{20.0} \Rightarrow C_{eq} = \mathbf{6.05\ \mu F}$

Example 14.4.5

All the capacitors are identical, each 1 μF. Find the equivalent capacitance between X and Y.



Solution

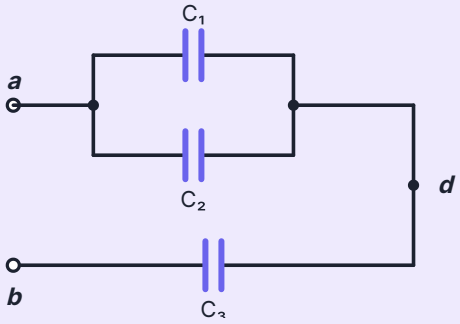
The three capacitors on the left (top-left, left side, bottom-left) are in series: C_a = 1/3 = 0.33 μF

C_a is in series with the top-right and bottom-right capacitors: $\frac{1}{C_b}$ = 1 + 1 + 3 ⇒ C_b = 0.20 μF

C_b is in parallel with the capacitor across XY: C_{eq} = 0.20 + 1 = **1.20 μF**

Exercise: combinations

1. C_1 = 2.00 μF, C_2 = 4.00 μF and C_3 = 9.00 μF; V_{ab} = 61.5 V. Find (a) the charge on each capacitor, (b) the p.d. across each, (c) the p.d. between a and d.



Answer

(a) Q_1 = 73.8 μC, Q_2 = 148 μC, Q_3 = 221 μC (b) V_1 = V_2 = 36.9 V, V_3 = 24.6 V (c) V_{ad} = 36.9 V

FULL SOLUTION

C_1 and C_2 are in parallel (between a and d); this combination is in series with C_3 (between d and b).

C_{12} = 2.00 + 4.00 = 6.00 μF

C_{eq} = $C_{12}C_3/(C_{12} + C_3)$ = (6.00)(9.00)/15.0 = 3.60 μF

Total charge Q = $C_{eq}V_{ab}$ = 3.60 × 61.5 = 221 μC (this is the charge on C_3 and on the C_{12} combination)

(b) $V_3 = Q/C_3$ = 221.4 / 9.00 = **24.6 V**

$V_1 = V_2 = Q/C_{12}$ = 221.4 / 6.00 = **36.9 V**

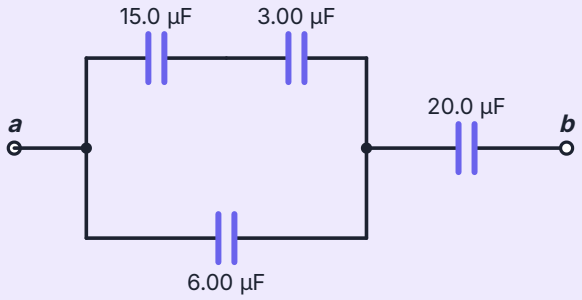
(a) $Q_1 = C_1V_1$ = 2.00 × 36.9 = **73.8 μC**

$Q_2 = C_2V_2$ = 4.00 × 36.9 = **148 μC**

Q_3 = **221 μC**

(c) $V_{ad} = V_1$ = **36.9 V**

2. Find (a) the equivalent capacitance between a and b, (b) the charge on each capacitor if V_{ab} = 15.0 V.



Answer

(a) 5.96 μF (b) 89.5 μC on 20 μF, 63.2 μC on 6 μF, 26.3 μC on both 15 μF and 3 μF

FULL SOLUTION

15.0 μF and 3.00 μF are in series; that branch is in parallel with 6.00 μF; the combination is in series with 20.0 μF.

(a) Series pair: (15.0)(3.00)/(15.0 + 3.00) = 2.50 μF

Parallel with 6.00 μF: 2.50 + 6.00 = 8.50 μF

In series with 20.0 μF: C_{ab} = (8.50)(20.0)/(8.50 + 20.0)

= **5.96 μF**

(b) $Q = C_{ab}V_{ab}$ = 5.965 × 15.0 = 89.5 μC

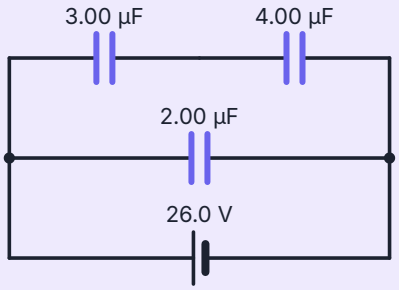
Charge on 20.0 μF = **89.5 μC**

P.d. across the parallel part = 89.5 / 8.50 = 10.53 V

Charge on 6.00 μF = 6.00 × 10.53 = **63.2 μC**

Charge on 15.0 μF and on 3.00 μF (series, same charge) = 2.50 × 10.53 = **26.3 μC**

3. A 3.00 μF and a 4.00 μF capacitor in series are connected in parallel with a 2.00 μF capacitor, across 26.0 V. Find (a) the equivalent capacitance, (b) the p.d. across each capacitor.



Answer

(a) 3.71 μF (b) 2.00 μF: 26.0 V; 3.00 μF: 14.9 V; 4.00 μF: 11.1 V

FULL SOLUTION

(a) 3.00 μF and 4.00 μF in series: (3.00)(4.00)/7.00 = 1.714 μF

In parallel with 2.00 μF: C_{eq} = 1.714 + 2.00

= **3.71 μF**

(b) The 2.00 μF capacitor is directly across the supply: V = **26.0 V**

Series branch: Q = 1.714 × 26.0 = 44.6 μC

V_3 = 44.6 / 3.00 = **14.9 V**; V_4 = 44.6 / 4.00 = **11.1 V** (check: 14.9 + 11.1 = 26.0 V)

14.5 Energy stored in a charged capacitor

- Work must be done to charge a capacitor: once some charge is on the plates, adding more charge of the same sign needs work against the repulsion. This work is stored as electric potential energy in the field between the plates.
- On a V–Q graph (a straight line, since $V = Q/C$), the energy stored is the **area under the graph**.

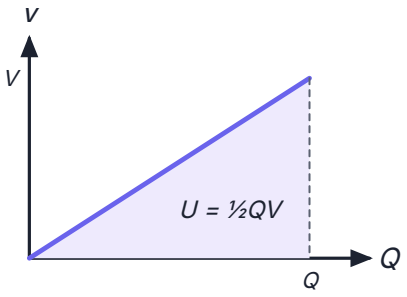


Figure 14.6: the energy stored is the area of the triangle under the V–Q graph.

$$U = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$$

unit: joule. Derivations are not required by the syllabus.

For understanding: moving a small charge dq through the p.d. $V = q/C$ needs d$W = V\,\mathrm{d}q$, so $W = \int_0^Q (q/C)\,\mathrm{d}q = Q^2/2C$.

Teacher's note

A battery that charges a capacitor supplies energy QV, but the capacitor stores only $\frac{1}{2}QV$. The other half is changed into heat in the resistance of the circuit (and some electromagnetic radiation), whatever the value of the resistance.

Example 14.5.1

A camera flash unit stores energy in a 150 μF capacitor at 200 V. How much energy is stored?

Solution

$U = \frac{1}{2}CV^2 = \frac{1}{2}(150 \times 10^{-6})(200)^2 =$ **3.0 J**

Example 14.5.2

A 2 μF capacitor is charged to 200 V by a battery. Find (a) the charge delivered by the battery, (b) the energy supplied by the battery, (c) the energy stored in the capacitor.

Solution

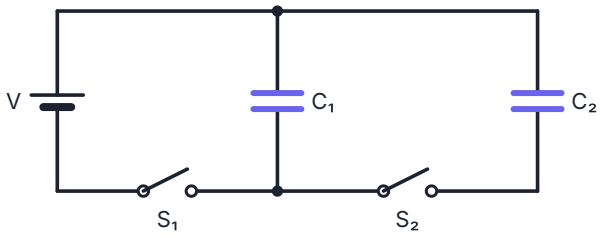
(a) $Q = CV = (2 \times 10^{-6})(200) =$ **4.0×10^{-4} C**

(b) $E = QV = (4.0 \times 10^{-4})(200) =$ **0.080 J**

(c) $U = \frac{1}{2}QV =$ **0.040 J**: half of the energy supplied

Example 14.5.3

C_1 = 6.00 μF is charged by closing S_1 (V = 20.0 V). S_1 is then opened and S_2 closed, connecting C_1 to the uncharged C_2 = 3.00 μF. Find (a) the initial charge on C_1, (b) the final p.d., (c) the final charge on each capacitor, (d) the energy lost.



Solution

(a) $Q_1 = C_1V$ = (6.00 μF)(20.0) = **120 μC**

(b) After S_2 is closed the capacitors are in parallel and share the charge until their p.d.s are equal. Charge is conserved: $V' = \dfrac{Q_{\text{total}}}{C_1 + C_2} = \dfrac{120}{9.00}$ = **13.3 V**

(c) Q_1' = (6.00)(13.33) = **80.0 μC**; Q_2' = (3.00)(13.33) = **40.0 μC**

(d) before: ½(120 × 10^{-6})(20.0) = 1.20 × 10^{-3} J; after: ½(120 × 10^{-6})(13.33) = 0.80 × 10^{-3} J; lost = **4.0 × 10^{-4} J** (as heat in the connecting wires)

Store slowly, release quickly: camera flash and defibrillator

A capacitor can be charged slowly by a small battery and then discharged in a fraction of a millisecond. The **same energy** released in a much **shorter time** gives a huge **power**. This is how a camera flash and a heart defibrillator work.

- **Charging** goes through a large resistance (the charging circuit), so the time constant RC is about a second or more: you wait for the "ready" light.
- **Discharging** goes through a small resistance (the flash tube, or the patient's chest, about 50 Ω), so the time constant is about a millisecond.
- Energy stored: U = ½CV^2. Power ≈ energy ÷ time, so releasing it in about a millisecond gives tens of kilowatts.

Simulation Camera flash and defibrillator

Choose the device and press Charge: watch the capacitor charge slowly (seconds). Then press Fire: the energy comes out in about a millisecond. A defibrillator shock stops the chaotic heartbeat so that a normal rhythm can restart.







capacitor	energy stored ½CV²	charging time constant RC
32 µF charged to 3.5 kV	**196 J**	**3.2 s**

discharge time constant RC	peak current V/R	power ≈ U ÷ 2RC
1.60 ms	**70 A**	**61 kW**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 14 to use the simulation.

Example 14.5.4

A defibrillator has a 32 μF capacitor charged to 3.5 kV. It discharges through the patient's chest, of resistance 50 Ω. Find (a) the energy stored, (b) the time constant of the discharge, (c) the initial current, (d) the average power if the energy is delivered in 3.0 ms.

Solution

(a) $U = \tfrac{1}{2}CV^2 = \tfrac{1}{2}(32 \times 10^{-6})(3500)^2 =$ **196 J**

(b) $\tau = RC = (50)(32 \times 10^{-6}) =$ **1.6×10^{-3} s**

(c) $I_0 = \dfrac{V_0}{R} = \dfrac{3500}{50} =$ **70 A**

(d) $P \approx \dfrac{196}{3.0 \times 10^{-3}} =$ **6.5×10^4 W** (65 kW): far more than its battery could give directly

Exercise: energy

1. C_1 = 3.00 μF and C_2 = 6.00 μF are connected in series to a 4.00 V battery. Find (a) the equivalent capacitance, (b) the charge on each capacitor, (c) the p.d. across each, (d) the energy stored in each, (e) the area of each plate of C_1 if its plates are 0.010 mm apart in vacuum.

Answer

(a) 2.00 μF (b) 8.00 μC (c) V_1 = 2.67 V, V_2 = 1.33 V (d) $U_1 = 1.07 \times 10^{-5}$ J, $U_2 = 5.33 \times 10^{-6}$ J (e) 3.39 m²

FULL SOLUTION

(a) Series: $C_{eq} = C_1C_2/(C_1 + C_2) = (3.00)(6.00)/9.00$

= **2.00 μF**

(b) $Q = C_{eq}V = 2.00 \times 4.00$

= **8.00 μC** on each capacitor (series)

(c) $V_1 = Q/C_1 = 8.00 / 3.00 =$ **2.67 V**; $V_2 = 8.00 / 6.00 =$ **1.33 V**

(d) $U = \tfrac{1}{2}QV$: $U_1 = \tfrac{1}{2}(8.00 \times 10^{-6})(2.67) =$ **1.07×10^{-5} J**

$U_2 = \tfrac{1}{2}(8.00 \times 10^{-6})(1.33) =$ **5.33×10^{-6} J**

(e) $C_1 = \varepsilon_0 A/d \Rightarrow A = C_1 d/\varepsilon_0 = (3.00 \times 10^{-6})(0.010 \times 10^{-3}) / (8.85 \times 10^{-12})$

= **3.39 m²**

14.6 Charging and discharging through a resistor

Charging

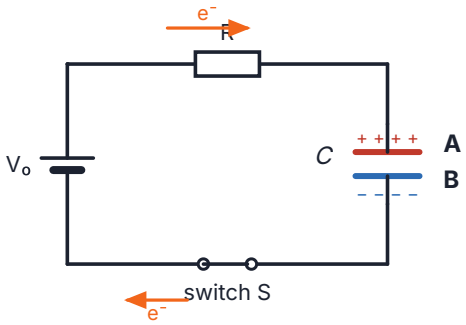


Figure 14.7: charging a capacitor through a resistor.

- When S is closed, electrons flow from the negative terminal of the battery, through R, onto plate B. Electrons leave plate A for the positive terminal, so A becomes positive.
- At the start the capacitor is uncharged, so the whole e.m.f. is across R: the current is largest, $I_0 = V_0/R$.
- As charge builds up, the p.d. across C rises, so the p.d. across R (and the current) falls.
- Charging stops when the p.d. across C equals V_0: then $I = 0$ and the charge is maximum, $Q_0 = CV_0$.

Derivation (charging)

Kirchhoff's loop rule at any time: $V_0 = IR + \dfrac{Q}{C}$, with $I = \dfrac{\mathrm{d}Q}{\mathrm{d}t}$

Rearrange: $\dfrac{\mathrm{d}Q}{\mathrm{d}t} = \dfrac{CV_0 - Q}{RC} = \dfrac{Q_0 - Q}{RC} \Rightarrow \int_0^{Q} \dfrac{\mathrm{d}Q}{Q_0 - Q} = \int_0^{t} \dfrac{\mathrm{d}t}{RC}$

$-\ln \dfrac{Q_0 - Q}{Q_0} = \dfrac{t}{RC} \Rightarrow Q = Q_0(1 - e^{-t/RC})$

Divide by C: $V = V_0(1 - e^{-t/RC})$. Differentiate: $I = \dfrac{\mathrm{d}Q}{\mathrm{d}t} = \dfrac{Q_0}{RC} e^{-t/RC} = I_0 e^{-t/RC}$

$$Q = Q_0(1 - e^{-t/RC}) \quad V = V_0(1 - e^{-t/RC}) \quad I = I_0 e^{-t/RC}$$

charging: $Q_0 = CV_0$, $I_0 = V_0/R$



Figure 14.8: charging. Q and V rise to their maximum; the current falls exponentially.

Discharging

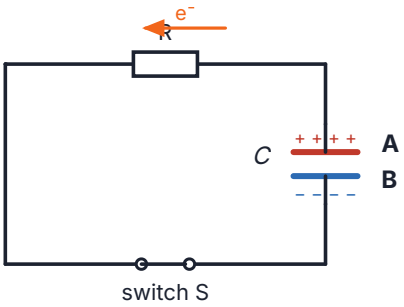


Figure 14.9: discharging a charged capacitor through a resistor.

- When S is closed, electrons flow from plate B through R to plate A and neutralise its positive charge.
- At the start the p.d. is V_0, so the current is largest, $I_0 = V_0/R$.
- As the charge falls, the p.d. falls, so the current falls. Discharging ends when $Q = 0$, $V = 0$ and $I = 0$.
- The current flows in the opposite direction to the charging current.

Derivation (discharging)

Loop rule with no battery: $0 = IR + \frac{Q}{C}$, with $I = \frac{dQ}{dt}$ (the charge decreases)

$$\frac{dQ}{Q} = -\frac{dt}{RC} \Rightarrow \ln\frac{Q}{Q_0} = -\frac{t}{RC} \Rightarrow Q = Q_0 e^{-t/RC}$$

$V = Q/C = V_0 e^{-t/RC}$; $I = \frac{dQ}{dt} = -I_0 e^{-t/RC}$ (minus: opposite direction; use the size in calculations)

$$Q = Q_0 e^{-t/RC} \quad V = V_0 e^{-t/RC} \quad I = I_0 e^{-t/RC}$$

discharging



Figure 14.10: discharging. Q, V and I all fall exponentially.

Time constant

Time constant, τ

of an RC circuit: the **time for the charge (or current) to fall to 1/e (37%) of its initial value** during discharging. During charging, it is the time for the charge to rise to (1 – 1/e), 63% of its final value.

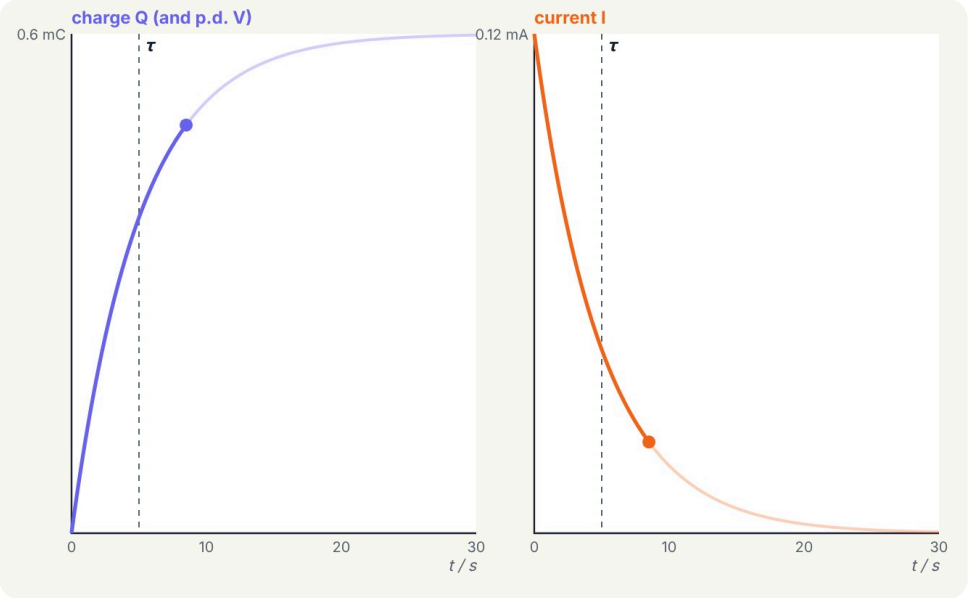
$$\tau = RC$$

unit: second (Ω × F = s). A large τ means slow charging and discharging.

- At t = τ: e^{-1} = 0.37, so discharging gives Q = 0.37Q_0; charging gives Q = 0.63Q_0 and I = 0.37I_0.
- After about 5τ the capacitor is practically fully charged (99.3%) or fully discharged.
- Half-life: the time for Q to halve is $t_{½}$ = RC ln 2 = 0.693RC.

Simulation — Charging and discharging through a resistor

Choose charging or discharging, then change R and C. Watch how the time constant τ = RC sets how fast the curves rise and fall. The dashed line marks t = τ.



time constant τ = RC	t	charge Q	p.d. across C
5.0 s	**8.5 s (= 1.70τ)**	**0.49 mC (82% of Q_0)**	**9.81 V**

current I	supply
0.022 mA (18% of I_0)	**12 V battery**

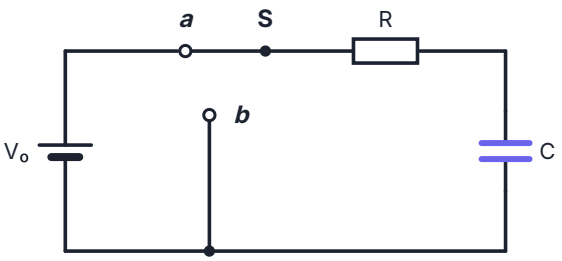
This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 14 to use the simulation.

Teacher's note

To find a time, take natural logs: e.g. discharging, $\frac{Q}{Q_0} = e^{-t/RC} \Rightarrow t = -RC \ln \frac{Q}{Q_0} = RC \ln \frac{Q_0}{Q}$.

Example 14.6.1

A capacitor is fully charged by a 20.0 V battery, then at t = 0 the switch is moved from a to b. C = 1.02 μF and the current falls to half its initial value in 40 μs. Find (a) R, (b) the time constant, (c) the charge at t = 0, (d) the charge at t = 60 μs.



Solution

(a) $0.50 = e^{-40 \times 10^{-6}/RC} \Rightarrow RC = \dfrac{40 \times 10^{-6}}{\ln 2} = 5.77 \times 10^{-5}$ s $\Rightarrow R = \dfrac{5.77 \times 10^{-5}}{1.02 \times 10^{-6}} =$ **57 Ω**

(b) $\tau = RC =$ **5.8×10^{-5} s**

(c) $Q_0 = CV_0 = (1.02 \times 10^{-6})(20.0) =$ **2.04×10^{-5} C**

(d) $Q = (2.04 \times 10^{-5})e^{-60/57.7} =$ **7.2×10^{-6} C**

Example 14.6.2

A 470 μF capacitor is charged through a 100 kΩ resistor from a 9.0 V supply. Find (a) the time constant, (b) the initial current, (c) the p.d. across the capacitor after 30 s, (d) the time for the capacitor to reach 90% of its final charge.

Solution

(a) $\tau = RC = (100 \times 10^3)(470 \times 10^{-6}) =$ **47 s**

(b) $I_0 = V_0/R = 9.0/100 \times 10^3 =$ **9.0×10^{-5} A**

(c) $V = 9.0(1 - e^{-30/47}) = 9.0(1 - 0.528) =$ **4.2 V**

(d) $0.90 = 1 - e^{-t/47} \Rightarrow t = 47 \ln 10 =$ **108 s** (2.3τ)

Exercise: RC circuits

1. A 1000 μF capacitor charged to 12 V is discharged through a 10 kΩ resistor. Find (a) the time constant, (b) the initial current, (c) the p.d. after 5.0 s, (d) the time for the charge to halve.

   Answer

   (a) 10 s (b) 1.2 mA (c) $12e^{-0.5} = 7.3$ V (d) $10 \ln 2 = 6.9$ s

   FULL SOLUTION

   C = 1000 μF = 1.0×10^{-3} F, R = 10 kΩ = 1.0×10^4 Ω, V_0 = 12 V

   (a) $\tau = RC = (1.0 \times 10^4)(1.0 \times 10^{-3})$

   = **10 s**

   (b) $I_0 = V_0/R = 12 / (1.0 \times 10^4)$

   = 1.2×10^{-3} A = **1.2 mA**

   (c) $V = V_0e^{-t/\tau} = 12e^{-5.0/10} = 12e^{-0.5}$

   = **7.3 V**

   (d) $Q = Q_0e^{-t/\tau} = \frac{1}{2}Q_0 \Rightarrow e^{-t/\tau} = \frac{1}{2} \Rightarrow t = \tau \ln 2 = 10 \times 0.693$

   = **6.9 s**

2. Sketch, on the same axes, the current–time graphs for charging the same capacitor through (a) a resistor R, (b) a resistor $2R$, from the same supply.

   Answer

   (a) starts at $I_0 = V_0/R$ and falls with $\tau = RC$. (b) starts at half the value, $I_0/2$, and falls more slowly ($\tau = 2RC$). The areas under both graphs are equal (the same final charge $Q_0 = CV_0$).

FULL SOLUTION

Charging current: $I = I_0 e^{-t/RC}$, with $I_0 = V_0/R$.

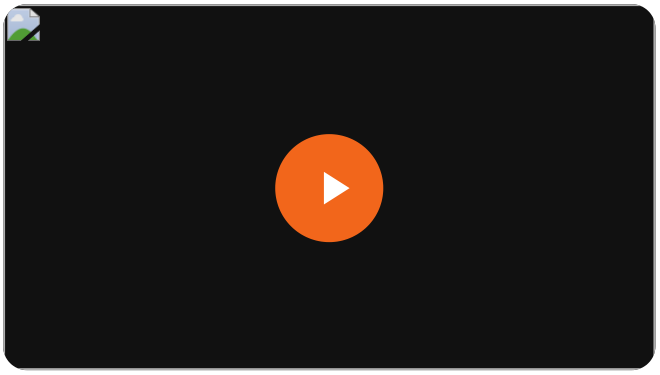
(a) With R: the curve **starts at $I_0 = V_0/R$** and decays exponentially with time constant $\tau = RC$.

(b) With 2R: the curve **starts at $I_0/2$** and decays **more slowly** ($\tau = 2RC$), so it lies above curve (a) at later times.

The area under an I–t graph is the charge. Both end with the same charge $Q_0 = CV_0$, so **the areas under the two graphs are equal**.

Physics in Real Life

Real-world videos for this chapter. Tap a picture to play it here.



WORKED EXAMPLE

Example: A Defibrillator is a Capacitor

Open in YouTube ↗

A defibrillator is a capacitor

The defibrillator charges a capacitor to a high voltage, then discharges its stored energy, $U = \frac{1}{2}CV^2$, through the patient's chest in a few milliseconds.